

Irrationality of Cubic Threefolds after One Stabilization

Tavis Rudd

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Abstract

We prove that $X \times \mathbf{P}^1$ is irrational for every smooth complex cubic threefold X . We define an integer-valued count from the formal exponents of certain rank-two summands of the quantum D -module. Iritani's blowup decomposition and weak factorization make this count birationally invariant for smooth projective fourfolds: every point, curve, and surface center contributes zero to the blowup formula. The count is one on a cubic threefold. Iritani–Koto's projective-bundle decomposition shows that it is two on $X \times \mathbf{P}^1$, whereas it is zero on $\mathbf{P}^4$. Applications include prime Fano threefolds of genus eight and universally CH_0 -trivial irrational fourfolds. The count also extends to an additive invariant of the Grothendieck group of varieties.

Keywords. cubic threefold; universal CH_0 -triviality; irrationality; quantum D -modules; formal exponents; birational invariants; weak factorization; Fano threefolds.

1 Introduction

Let X be a smooth complex cubic threefold. Clemens and Griffiths proved that X is irrational [8], but their intermediate-Jacobian argument does not directly obstruct the rationality of $X \times \mathbf{P}^1$: in a weak factorization of a birational map between smooth fourfolds, the blowup formula for a surface center S contributes $H^3(S)(-1)$ in degree five, while $H^3(X)(-1)$ occurs in $H^5(X \times \mathbf{P}^1)$. We prove that the stabilized fourfold is nevertheless irrational.

Theorem 1.1 (One-step irrationality). *For every smooth complex cubic threefold X , the fourfold $X \times \mathbf{P}^1$ is irrational.*

The birational argument is short. We construct an integer-valued *exponent count* I_{exp} with

$$I_{\text{exp}}(X) = 1, \quad I_{\text{exp}}(X \times \mathbf{P}^1) = 2, \quad I_{\text{exp}}(\mathbf{P}^4) = 0.$$

For a blowup along a smooth center of codimension c , its change is

$$I_{\text{exp}}(\text{Bl}_Z Y) = I_{\text{exp}}(Y) + \sum_{j=0}^{c-2} c_{\text{exp}}(\omega_j).$$

Here each ω_j records one center summand in the quantum blowup comparison. Every such contribution vanishes when $\dim Z \leq 2$. Weak factorization of smooth projective fourfolds uses only these centers, so the count is birationally invariant in dimension four. Its values above rule out a birational map $X \times \mathbf{P}^1 \dashrightarrow \mathbf{P}^4$. Section 2 constructs the count and proves these properties. This also explains the dimensional limit: a factorization in dimension five may use threefold centers, whose contributions need not vanish.

The stable-rationality picture varies across the moduli space. Hassett and Tschinkel identified varieties birational to cubic threefolds as the exceptional Fano-threefold case not covered by their stable-irrationality theorem [11]. More recently, Engel, de Gaay Fortman, and Schreieder proved that a very general complex cubic threefold is not stably rational [10, Corollary 1.4], whereas Tschinkel and Zhang constructed explicit stably rational smooth cubic threefolds over $\mathbf{Q}$ [21, Theorem 1.3]. The theorem above applies uniformly in both situations: it also proves that the first stabilization of those stably rational examples is irrational. A companion paper proves that this bound is sharp: it gives two explicit examples for which $X \times \mathbf{P}^2$ is rational [20].

The proof uses the quantum D -module (QDM): the cohomology module over the ring of Novikov and bulk parameters, equipped with its flat quantum connection [16]. The Novikov parameters record curve classes, while the bulk parameters are coordinates on cohomology. By the even QDM we mean the restriction to even cohomology with the odd bulk variables set to zero. We then replace the ring of numerical Novikov and even bulk parameters by its fraction field; this is what *generic* means below. In particular, it refers to quantum parameters, not to a general point in the moduli space of cubic threefolds. In the direction of the connection variable z , the leading coefficient is quantum multiplication by the Euler field. Over an algebraic closure of the generic coefficient field, its generalized eigenspaces decompose the formal connection into blocks.

The count selects rank-two blocks for which the centered leading term N of Euler multiplication is nonzero and satisfies $N^2 = 0$. Horizontality of the Poincaré pairing makes a canonical elementary modification regular singular, while flatness gives a Lax equation for its residue, whose eigenvalues are therefore preserved. We count the block when the two residue eigenvalues are distinct modulo $\mathbf{Z}$, equivalently when the two formal exponent classes are distinct. Using Beauville’s quantum product formulas, we compute the exponent representatives $-1/6$ and $-5/6$ for the cubic block, with difference $2/3$. The resulting invariant vanishes on points, curves, and surfaces and on projective space. Iritani–Koto’s projective-bundle theorem shows that it doubles after product with $\mathbf{P}^1$. This proves the theorem.

This use of Euler eigenspace blocks in birational geometry belongs to the same circle of ideas as the Hodge-atom framework of Katzarkov–Kontsevich–Pantev–Yu [17]. Their framework combines the generalized eigenspaces of Euler multiplication, Iritani’s blowup theorem, and weak factorization to produce birational invariants. In particular, their Example 6.21 uses the zero spectral piece, enhanced by the Serre operator and formal monodromy, to recover the irrationality of every smooth cubic threefold [17, Example 6.21]. Recent applications and refinements include work on cubic, Gushel–Mukai, and Khchle fourfolds [12, 6, 5].

The invariant used here retains less structure: it uses no Hodge structure and records only the rank-two formal-exponent condition above. Its purpose is the one-stabilization problem. For this purpose, we specialize Iritani’s blowup comparison in a way that keeps the center summands distinct, prove that the invariant vanishes on points, curves, and surfaces, and use Iritani–Koto’s projective-bundle comparison for stabilization [14, 15]. Gyenge gives a complementary formal tensor-product theorem for quantum D -modules of products and proves multiplicativity for Hodge atoms [13]. Cai independently computes the same two exponent representatives and uses them with Iritani’s blowup theorem to prove symplectic irrationality of cubic threefolds [7]; that argument does not address stabilization.

The same argument applies to any smooth projective threefold with positive exponent count. It also gives irrationality after one stabilization for prime Fano threefolds of genus eight, and universally CH_0 -trivial irrational fourfolds. These consequences are stated and proved in Section 3. The additive and spectral extensions in Section 4 follow the proof of

the cubic theorem and its applications.

2 Quantum connections and one stabilization

We prove the theorem by applying the quantum D -module decomposition formulas directly. The common argument is stated first for an arbitrary additive invariant. We then construct the specific invariant used for a cubic threefold and prove that it vanishes on points, curves, and surfaces.

Here is the concrete invariant before the general construction. Split the generic even quantum D -module into formal connection blocks. A block contributes one if it has rank two, its centered leading term N is nonzero with $N^2 = 0$, and the two residue eigenvalues after the canonical elementary modification are distinct modulo $\mathbf{Z}$. These eigenvalues represent the block's two formal exponent classes. Every other block contributes zero, and the sum of the contributions is I_{exp} . The rest of the section shows that this count is intrinsic and additive, vanishes on points, curves, and surfaces, equals one on a cubic threefold, doubles after product with $\mathbf{P}^1$, and vanishes on $\mathbf{P}^4$.

2.1 Block multisets and a single blowup

Let Y be smooth and projective. We first make three setup choices. Pass to numerical Novikov variables, set the odd bulk variables to zero, and restrict the underlying cohomology space to $H^{\text{ev}}(Y)$. The last restriction is essential: restricting the bulk base alone would still leave odd cohomology in the fibre. Use graded completions as in [14, Section 2.2], and let K_Y be the fraction field of the numerical divisor-equation-reduced coefficient domain described below. Fix an algebraic closure $\overline{K}_Y$. We form all spectral blocks only after base change to $\overline{K}_Y$, so no later scalar extension can split a factor that has already been counted as one block.

For a homogeneous even cohomology basis (ϕ_α) , our conventions are

$$\nabla_{\partial_t \alpha} = \partial_t \alpha + z^{-1}(\phi_\alpha \star_t), \quad \nabla_{z \partial_z} = z \partial_z - z^{-1}(E_Y \star_t) + \mu_Y,$$

where $n = \dim Y$, $\mu_Y|_{H^k(Y)} = (k - n)/2$, and

$$E_Y = c_1(Y) + \sum_{\alpha} \left(1 - \frac{\deg \phi_\alpha}{2}\right) t^\alpha \phi_\alpha.$$

Horizontal sections therefore satisfy $z^2 \partial_z y = ((E_Y \star_t) - z \mu_Y) y$. Let U denote multiplication by the Euler field on $H^{\text{ev}}(Y)$. For a primary block with eigenvalue λ , centering removes the scalar exponential factor $e^{-\lambda/z}$; its leading operator becomes $N = U - \lambda I$.

After base change to $\overline{K}_Y$, each generalized eigenspace of U is a summand of the full formal connection, not just an invariant subspace of U :

Proposition 2.1 (Formal spectral connection splitting). *Suppose the leading operator of a finite free formal QDM has block-diagonal form $U_i = \lambda_i I + N_i$, where each N_i is nilpotent and $\lambda_i - \lambda_j$ is a unit for $i \neq j$. There is a unique gauge $G = I + O(z)$, regular in z and block off-diagonal in positive orders, which makes the z -connection block diagonal. In this gauge every bulk and Novikov direction is block diagonal as well. The horizontal pairing is block diagonal and nondegenerate on each factor.*

Proof. At order z^{n-1} , the (i, j) -entry of the gauge equation is a Sylvester equation

$$U_i X - X U_j = B_{n,ij}.$$

Its operator is $(\lambda_i - \lambda_j)I$ plus the nilpotent operator $X \mapsto N_i X - X N_j$, hence is invertible. Recursion gives the normalized gauge, and the same equation with zero right-hand side proves uniqueness.

Choose frames of the generalized-eigenspace subbundles after the splitting extension. If an off-diagonal block of a base-direction connection had a first nonzero Laurent coefficient Xz^m , the lowest off-diagonal coefficient of flatness would be $U_i X - X U_j = 0$, contrary to Sylvester invertibility. Thus every base direction preserves the same factors. Applied coefficientwise to pairing horizontality, the identical argument kills the off-diagonal pairing blocks. Nondegeneracy of the total pairing then makes every diagonal restriction nondegenerate. $\square$

We call these factors the *generic even QDM blocks* of Y . Each is a whole primary spectral summand; individual Jordan blocks within one primary summand are not counted separately. A block retains its connection, pairing, and centered Euler endomorphism, but not an absolute cohomological grading. A block type may retain additional marked data. If it retains grading, a morphism may carry a recorded integral suspension, and the invariant must be invariant under that suspension. The block type used below forgets absolute grading. Its morphisms preserve the retained data and are regular connection isomorphisms after scalar extension and formal change of bulk coordinates. In particular, their gauges use neither negative nor fractional powers of the connection variable z .

Fix the retained data $\mathcal{T}$. Write $\mathcal{L}_{\mathcal{T}}$ for the monoid of finite multisets of regular-isomorphism classes of blocks with these data. Its addition is multiset union. Direct sum gives each Y its block multiset $\mathcal{B}_{\mathcal{T}}(Y) \in \mathcal{L}_{\mathcal{T}}$. An *additive invariant* is a commutative-monoid homomorphism

$$w : \mathcal{L}_{\mathcal{T}} \longrightarrow A, \quad I_w(Y) := w(\mathcal{B}_{\mathcal{T}}(Y)),$$

where A is a commutative monoid. This formulation also allows values in monoids without subtraction. The additive group-valued extension of the exponent count is treated after the main theorem.

We use the following notation.

Symbol	Meaning
$\mathcal{L}_{\mathcal{T}}$	finite multisets of regular-isomorphism classes of blocks
$\mathcal{B}_{\mathcal{T}}(Y)$	multiset of block isomorphism classes for Y
$w : \mathcal{L}_{\mathcal{T}} \rightarrow A$	additive invariant with values in A
ω_j	data attached to the j -th center summand
$c_w(\omega_j)$	invariant contribution of that center summand

Table 1: Notation for the block monoid and center summands.

Consider one blowup $\beta : \tilde{Y} = \text{Bl}_Z Y \rightarrow Y$ along a smooth center of codimension $c \geq 2$. Iritani's comparison has one ambient summand and $c - 1$ center summands. Distinct copies of the same center must remain distinct in the blowup formula. We therefore retain separate data for each center summand: for the j -th, record

$$\omega_j = (\beta, Z, j, \varsigma_j, \chi_j), \quad 0 \leq j \leq c - 2,$$

where ς_j is Iritani's center reconstruction parameter and χ_j is the induced numerical Novikov specialization. Its correction is $c_w(\omega_j) := w(\mathcal{B}_{\mathcal{T}}(Z; \omega_j))$. The comparison will give equality only after applying w ; distinct block multisets may have the same invariant.

Theorem 2.2 (A birational invariance criterion). *Let $w : \mathcal{L}_{\mathcal{T}} \rightarrow A$ be an additive invariant unchanged by the allowed regular scalar extensions and formal coordinate changes. Suppose*

$$I_w(\mathbf{P}_Y(V)) = rI_w(Y), \quad \text{rk } V = r, \quad (2.1)$$

$$I_w(\text{Bl}_Z Y) = I_w(Y) + \sum_{j=0}^{c-2} c_w(\omega_j), \quad \text{codim}_Y Z = c \geq 2. \quad (2.2)$$

If every summand associated with a smooth center of dimension at most $d - 2$ has zero correction, then I_w is a birational invariant of smooth projective d -folds.

Proof. Projective weak factorization writes a birational map of smooth projective d -folds as a zigzag of blowups and blowdowns along smooth centers of codimension at least two [1, Theorem 0.3.1]. By (2.2) and the vanishing of every center correction, the invariant is unchanged at each step; the equalities therefore telescope. Disconnected centers are treated componentwise, while a divisor blowup is an isomorphism. $\square$

Equivalently, the invariant descends on object sets:

$$\begin{array}{ccc} \text{Ob}(\text{SmProj}_d) & \xrightarrow{I_w} & A \\ \downarrow & & \parallel \\ \text{Ob}(\text{SmProj}_d)/\sim_{\text{bir}} & \xrightarrow{\bar{I}_w} & A. \end{array} \quad (2.3)$$

This square reformulates the proof; it is not an additional input.

2.2 Compatibility with blowups and projective bundles

To use the cited decomposition theorems, we must check that they preserve all the data seen by the invariant. Four issues arise: coefficient fields, restriction to even cohomology, regularity in z , and faithful embedding of the center coefficients. The last issue is the delicate one: Iritani's center Novikov map may have a kernel, so the independent target coordinates must recover the missing numerical curve-class data.

Coefficient fields. The intrinsic ample-adic completion and the Laurent neighborhood of a fibre or exceptional cusp need not map to one another. Iritani's display (1.1) supplies the required common coefficient maps [14, (1.1)]; injectivity for each center summand is established below. For projective bundles the common source is $\mathbf{C}[z, q][[Q, \hat{\tau}]]$. On the blowup side the ambient term uses the corresponding graded source. A center term begins with its intrinsic reduced coefficient ring from Remark 2.3, maps to the possibly smaller image ring of Remark 5.6, and is then pulled to a separate target ring for that summand in the external direct sum. Embed the generic fields of the ambient common image and of these separate center images in a fixed algebraically closed overfield and form every primary block there. Base change after this splitting extension preserves the rank, nilpotent Jordan type, regular formal gauge, and formal exponent classes of each block; forming primary factors over a nonsplitting field before extension would not.

Restriction to even cohomology. The comparison is restricted to even cohomology, not merely to the even bulk base. The supermodule conventions and the pull-push and Fourier formulas in Iritani's Section 5.8 and Iritani–Koto's construction are parity-preserving. After the odd bulk variables are set to zero, the comparison and its inverse therefore restrict to mutually inverse maps on H^{ev} . This is the QDM on which the additive invariant is defined.

Regularity in z . The comparisons are regular in z . Iritani's Theorem 5.18 and Iritani–Koto's Theorem 5.1 are respectively defined over rings of the form

$$\mathbf{C}[z]((q_{\text{exc}}^{-1/s}))[[Q, \tilde{\tau}]], \quad \mathbf{C}[z]((q_{\text{fib}}^{-1/r'}))[[Q, \tilde{\tau}]],$$

where Iritani–Koto take $r' = r$ when $r - 1$ is even and $r' = 2r$ when $r - 1$ is odd. Here $s \in \{c - 1, 2(c - 1)\}$ is Iritani's parity-dependent ramification index from [14, (5.11)]. Their inverses use no negative powers of z ; see also [15, Remark 5.3]. The maps are homogeneous but need not have degree zero: the projective-bundle map has degree $-(r - 1)$, while Iritani's ambient and center components have degrees 0 and $-c$, respectively [15, Theorem 5.1(3)][14, Theorem 5.18(3)]. Because the block type used here does not retain absolute grading, these shifts do not change its component. For a graded refinement they are recorded as integral suspensions, and the operation formula applies only to an invariant under those suspensions.

Faithful center coordinates. Iritani's center map need not be injective [14, (5.15)]. After numerical reduction it has the form

$$Q_Z^d \mapsto Q^{i_* d} q_{\text{exc}}^{-\rho_Z \cdot d / (c-1)}. \quad (2.4)$$

Here $\rho_Z = c_1(N_{Z/Y})$. The raw map is not the coefficient map used below. Remark 2.3 gives the intrinsic divisor-equation-reduced coefficient ring, whose Novikov generators are the combined symbols

$$X_d = Q_Z^d \exp(\sigma^{(2)} \cdot d);$$

the divisor coordinates $\sigma^{(2)}$ are not separate elements of this ring. Remark 5.6 places its pushforward under the raw Novikov map over the image of that reduced ring [14, Remarks 2.3 and 5.6]. In the external direct sum of Section 5.8.2, by contrast, the variables $s_j \in H^*(Z)$ are independent coordinates on the common comparison target, and the center parameter is identified as $\varsigma_j = \varsigma_j^\circ + s_j$ [14, (5.47)–(5.48)]. The divisor components of these independent target coordinates retain the curve-class information lost by (2.4).

Lemma 2.3 (Faithful center base change). *Let $R_{Z,\text{src}}^{\text{red}}$ be the coefficient domain whose $\mathbf{C}[z]$ -extension is Iritani's divisor-equation-reduced source ring $\mathbf{C}[z][[Q_Z e^{\sigma^{(2)}}, \sigma']][\sigma^0]$. Let A_j be the numerical coefficient domain of the external direct sum in Section 5.8.2, with the independent target coordinates $t, s_0, \dots, s_{c-2}$ and with z omitted. The pullback determined by the center Novikov map and $\varsigma_j = \varsigma_j^\circ + s_j$ gives a continuous homomorphism*

$$\iota_j : R_{Z,\text{src}}^{\text{red}} \longrightarrow A_j.$$

On a combined divisor-equation generator it is

$$X_d = Q_Z^d \exp(\sigma^{(2)} \cdot d) \mapsto Q^{i_* d} q_{\text{exc}}^{-\rho_Z \cdot d / (c-1)} \exp((\varsigma_j^{\circ, (2)} + s_j^{(2)}) \cdot d). \quad (2.5)$$

This map is injective. In particular, it induces a field embedding

$$\text{Frac}(R_{Z,\text{src}}^{\text{red}}) \hookrightarrow \text{Frac}(A_j).$$

After this field extension, each center summand in Iritani's comparison is the faithful scalar extension, followed by the formal coordinate change ς_j , of the intrinsic generic numerical even QDM of Z .

Proof. All completions here use the graded convention of [14, Section 2.2]. The pullback is defined on the reduced ring by the String and Divisor Equations, as explained after (5.36) and (5.47) there. It need not be defined on the unrestricted bulk power-series ring. We prove injectivity using the first nonzero Novikov degree, retaining the full exponentials in the target divisor coordinates. Choose integral divisors $D_1, \dots, D_\rho$ whose pairings separate $N_1(Z)$, and use their dual coordinates in $s_j^{(2)}$. The variables s_j occur only in the target A_j , not as coefficients in $R_{Z, \text{src}}^{\text{red}}$. This asymmetry is essential.

Filter by an integral ample degree pulled back from Y . Every fibre of the untagged monomial map in a bounded piece is finite: the restriction of an ample divisor from Y is ample on Z , and its bounded slice of $\bar{N}E(Z)$ contains only finitely many lattice points. This filtration is respected by the target map because $H \cdot i_* d = (i^* H) \cdot d$, and the fixed shifts have nonnegative Novikov degree.

For a fixed curve class, the coefficient of an element of the reduced ring is polynomial in the nondivisor and unit coordinates. Indeed, an element has finitely many homogeneous degrees and bounded unit degree; every other even nondivisor coordinate has strictly negative degree. Thus only finitely many of its monomials occur at that curve class. This is where the reduced, graded source is essential.

Suppose a nonzero source element maps to zero and take its first nonzero Novikov-degree part. At that degree only the Novikov-degree-zero part of ς_j° enters. Translation by its nondivisor and unit components is an injective substitution on the polynomial coefficients, over the target Laurent coefficient field. Its divisor component multiplies each exponential by a nonzero scalar. Group the terms by their common untagged target monomial. On one nonzero group the remaining relation has the form

$$\sum_{d \in F} a_d \exp \left(\sum_{k=1}^{\rho} (D_k \cdot d) s_{j,k} \right) = 0, \quad a_d \neq 0,$$

over a characteristic-zero coefficient domain, including the remaining bulk coordinates, that does not contain the variables $s_{j,k}$. The exponent vectors are distinct. Choose an integral one-parameter restriction on which they remain distinct. Derivatives of orders $0, \dots, |F| - 1$ at the origin give a Vandermonde matrix, hence every a_d is zero, a contradiction. This proves injectivity on the full reduced source. We have not truncated the divisor exponentials by bulk order; doing so could lose the distinction between curve classes.

Remark 2.3 of [14] gives the intrinsic reduced source, and Remark 5.6 gives its image under the raw Novikov map (5.15). Although that map can be noninjective on the unrestricted source, it retains the independent $\sigma^{(2)}$ coordinates in its target. Its restriction to the reduced source is injective by the same argument, with zero shift. Equations (5.47)–(5.48) then pull that image into the external direct sum over A_j , where the variables s_j are independent, and identify the direct sum with the blowup module. Associativity of scalar extension therefore identifies the center summand with scalar extension directly along the injective composite ι_j . Passing to fraction fields is faithfully flat. Finally, the inverse of the combined formal coordinate change $(\tau, \varsigma_0, \dots, \varsigma_{c-2})$, supplied by Theorem 5.18(7) and Section 5.8.2, transports the same statement to the original blowup coordinates. $\square$

Proposition 2.4 (Marker formulas for the QDM comparisons). *Let an additive invariant be preserved by the regular scalar extensions and formal coordinate changes described above, and by every recorded grading suspension if its block type retains grading. The projective-bundle and blowup QDM comparisons on even cohomology then supply (2.1) and (2.2); the contribution of the j -th center summand is evaluated using its separate data ω_j .*

Proof. Projective bundles. For a rank- r bundle, Iritani–Koto’s Theorem 5.1 gives r copies of the base QDM after Laurent extension [15, Theorem 5.1]. Its invertible joint bulk Jacobian gives independent unit coordinates on the r target factors. In the j -th factor, changing its unit coordinate adds an independent scalar $u_j I$ to Euler multiplication. For two finite spectra, the resultant of their translated characteristic polynomials is a nonzero polynomial in $u_i - u_j$, with leading coefficient one. Hence the spectra of distinct factors are disjoint at the generic point: their primary blocks do not fuse when the direct sum is formed. Regularity, restriction to even cohomology, the fixed splitting field, and suspension invariance therefore carry the invariant blockwise across the comparison, and additivity gives (2.1). Tensoring the bundle by a line bundle, when needed for the global-generation convention, changes neither the projective bundle nor the intrinsic Novikov embedding [15, Remark 1.2 and Remark 5.2].

Blowups. For a codimension- c blowup, Iritani’s Theorem 5.18 gives one ambient QDM and $c - 1$ center QDMs [14, Theorem 5.18]. Parts (4), (6), and (7) give the comparison asymptotics, the leading reconstruction parameters, and the invertible combined bulk Jacobian, respectively. The independent unit coordinates supplied by part (7) give the same nonzero resultant argument, so the spectra of the ambient and individual center summands are pairwise disjoint generically. Regularity, restriction to even cohomology, the fixed splitting field, suspension invariance, and Lemma 2.3 for each center copy then give (2.2). Each comparison is evaluated over its own common coefficient ring; the proof never composes Laurent comparisons recursively. $\square$

The QDM comparisons need not identify the underlying elements of the block monoid. They identify their images under the additive invariant w , whose additivity is expressed by the square

$$\begin{array}{ccc} \mathcal{L}_{\mathcal{T}} \times \mathcal{L}_{\mathcal{T}} & \xrightarrow{+} & \mathcal{L}_{\mathcal{T}} \\ w \times w \downarrow & & \downarrow w \\ A \times A & \xrightarrow{+} & A. \end{array} \quad (2.6)$$

Proposition 2.4 supplies the resulting equalities in A . Its blowup statement also retains the data distinguishing the center summands, as required by Theorem 2.2.

2.3 The rank-two exponent count

We now specialize the general construction to $A = \mathbf{N}$. The block type $\mathcal{T}_{\text{exp}}$ retains the connection, pairing, and centered Euler endomorphism on H^{ev} , and forgets absolute grading. Let E be a generic even block with centered connection

$$z \partial_z y = A(z)y, \quad A(z) = z^{-1}N + A_0 + zA_1 + z^2A_2 + \cdots. \quad (2.7)$$

Assume $\text{rk } E = 2$, $N \neq 0$, and $N^2 = 0$. Then $L = \text{im } N = \ker N$ is a line. The following matrix calculation explains the modification. In a frame adapted to L , suppose that A_0 preserves this line, so that

$$N = \begin{pmatrix} 0 & \nu \\ 0 & 0 \end{pmatrix}, \quad A_0 = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}, \quad (A_1)_{21} = c.$$

Changing the frame by $S = \text{diag}(1, z)$ removes the pole and gives residue

$$R = \begin{pmatrix} a & \nu \\ c & d - 1 \end{pmatrix}.$$

The lower-left entry explains why the first-order coefficient A_1 is needed; the subtraction of one comes from differentiating the new frame. Eigenvalues $0, 1$ represent the same class modulo $\mathbf{Z}$ and do not qualify, whereas $0, 2/3$ represent distinct classes.

The canonical elementary modification along this line is

$$E^\sharp = \{s \in E : s \bmod z \in L\}. \quad (2.8)$$

For this modification to remove the pole, the regular coefficient A_0 must preserve L . Pairing horizontality supplies exactly that condition.

Lemma 2.5 (The regular coefficient preserves the nilpotent line). *Horizontality of the Poincaré pairing implies $A_0 L \subset L$.*

Proof. Separated spectral factors are orthogonal for the horizontal Poincaré pairing: coefficientwise horizontality kills every off-diagonal block by an invertible Sylvester operator. Hence the restriction to E is nondegenerate. Write it as $P(z) = P_0 + zP_1 + \dots$. Horizontality gives

$$z\partial_z P(z) + A(z)^T P(z) + P(z)A(-z) = 0.$$

The z^{-1} coefficient gives $N^T P_0 = P_0 N$. Thus L is isotropic and, in dimension two, $L^\perp = L$. The constant coefficient is

$$A_0^T P_0 + P_0 A_0 + N^T P_1 - P_1 N = 0.$$

Evaluating this identity on (x, x) , where $x \in L = \ker N$, gives $(A_0 x, x) = 0$. Hence $A_0 x \in L^\perp = L$. $\square$

Choose a frame with $L = \langle e_1 \rangle$ and $N e_2 = \nu e_1$, $\nu \neq 0$. The modified lattice has frame $e_1, z e_2$, so

$$A^\sharp = S^{-1} A S - S^{-1} z \partial_z S, \quad S = \text{diag}(1, z). \quad (2.9)$$

The term $z^{-1} N$ becomes regular. The only possible new pole is the E_{21} -coefficient of A_0 , which vanishes by Lemma 2.5. Thus $A^\sharp = R + zR_1 + \dots$. Define

$$\delta^\sharp(E) := (\text{tr } R)^2 - 4 \det R. \quad (2.10)$$

Proposition 2.6 (Invariance of the modified residue). *Under formal change of even bulk coordinates, the modified connection is regular in every base direction and*

$$\partial R = [G_\partial, R] \quad (2.11)$$

for a regular matrix G_∂ . Consequently $\partial(\delta^\sharp) = 0$. Regular connection isomorphisms preserve the domain and value of $\delta^\sharp$.

Proof. There are two steps. Flatness first preserves the nonzero square-zero orbit of N . After the elementary modification, it removes the possible base pole and gives the Lax equation for the residue. Write a centered base equation as $\partial y = (z^{-1} C_\partial + C_{\partial,0} + O(z))y$. The z^{-2} coefficient of flatness gives $[N, C_\partial] = 0$, and the trace of the z^{-1} coefficient gives $\text{tr } C_\partial = 0$; here the base equation has been scalar-centered. Nilpotence of N away from the base point is not yet known, so the commutant is computed over the complete local ring $\mathcal{O}$ of the formal bulk germ. The rank-two spectral cluster continues over $\mathcal{O}$ because it is separated from the complementary spectral clusters at the base point; N is the centered z^{-1} coefficient of that continuation, so $\text{tr } N = 0$, and it specializes at the closed point to a nonzero rank-one nilpotent. Choose v with v, Nv a basis modulo the maximal ideal; by Nakayama it is a basis over $\mathcal{O}$, so N is cyclic. Any C commuting with a cyclic N is

determined by $Cv = \alpha v + \beta Nv$ and equals $\alpha I + \beta N$, whether or not N is nilpotent. Hence $[N, C_\partial] = 0$ and $\text{tr } C_\partial = 0 = \text{tr } N$ give $C_\partial = q_\partial N$ with $q_\partial \in \mathcal{O}$. The same z^{-1} coefficient gives

$$\partial N = -q_\partial N + [N, q_\partial A_0 - C_{\partial,0}].$$

Writing $B = q_\partial A_0 - C_{\partial,0}$, one has $\partial(N^2) = -2q_\partial N^2 + [N^2, B]$; uniqueness for the formal coefficient recursion preserves $N^2 = 0$. An entry nonzero at the closed point remains a unit, so N has rank one throughout the formal germ. Thus formal bulk transport preserves the nonzero square-zero orbit and its image line. After modification, the only possible base pole is $z^{-1}kE_{21}$. If $R = \begin{pmatrix} a & \nu \\ c & d \end{pmatrix}$, the z^{-1} flatness equation is $kE_{21} + [R, kE_{21}] = 0$. Its diagonal entries are νk and $-\nu k$, so $k = 0$. The constant term of the flatness equation is now (2.11); trace and determinant are constant under infinitesimal conjugation. A regular isomorphism carries N , L , and the modified lattice to their counterparts and conjugates the residue. $\square$

Proposition 2.7 (Exponent interpretation). *If ρ_1, ρ_2 are the eigenvalues of R , their classes modulo $\mathbf{Z}$ are the formal exponent classes of the original centered block and*

$$\delta^\sharp = (\rho_1 - \rho_2)^2.$$

Proof. The elementary modification uses integral powers of z , so it preserves formal exponent classes modulo $\mathbf{Z}$. The modified connection is regular singular with residue R ; the regular-singular classification identifies the exponent classes with the residue eigenvalues modulo $\mathbf{Z}$ [19, Chapter 3]. The displayed identity is the quadratic discriminant formula. $\square$

For such a block E , write ρ_1, ρ_2 for the eigenvalues of its modified residue in the fixed algebraic closure $\overline{K}$ of the generic coefficient field. Define the invariant on each block by

$$w_{\text{exp}}(E) = \begin{cases} 1, & \text{rk } E = 2, N \neq 0, N^2 = 0, [\rho_1] \neq [\rho_2] \text{ in } \overline{K}/\mathbf{Z}, \\ 0, & \text{otherwise,} \end{cases} \quad (2.12)$$

and extend it additively to $w_{\text{exp}} : \mathcal{L}\mathcal{T}_{\text{exp}} \rightarrow \mathbf{N}$. Put $I_{\text{exp}}(Y) = w_{\text{exp}}(\mathcal{B}\mathcal{T}_{\text{exp}}(Y))$. Propositions 2.6 and 2.7 make this a well-defined additive invariant for Proposition 2.4: conjugation preserves the residue eigenvalues, and a common scalar shift preserves their difference. The use of exponent classes also makes the scalar test insensitive to integral changes of representatives. The condition $\delta^\sharp \neq 0$ is insufficient: it says only $\rho_1 \neq \rho_2$ and therefore also marks resonant blocks.

2.4 The cubic block

Only a rank-two block can contribute to I_{exp} . For a cubic threefold, the calculation below isolates the unique such candidate and then computes the residue of its elementary modification.

Let $X \subset \mathbf{P}^4$ be a smooth cubic threefold, let $P = H|_X$, and put $q = Q^\ell$ for the line class and $r = (3q)^{1/2}$. Beauville's formulas give [3, main theorem and (2.1)–(2.3)]

$$P \star P = P^2 + 6q, \quad P \star P^2 = P^3 + 15qP, \quad P \star P^3 = 6qP^2 + 36q^2. \quad (2.13)$$

Thus, in the basis $(1, P, P^2, P^3)$,

$$U_X = 2 \begin{pmatrix} 0 & 6q & 0 & 36q^2 \\ 1 & 0 & 15q & 0 \\ 0 & 1 & 0 & 6q \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad D_X = \frac{1}{2} \text{diag}(3, 1, -1, -3), \quad (2.14)$$

and $z^2 \partial_z y = (U_X + z D_X)y$. This agrees with [7, Section 3].

Proposition 2.8 (Small-even cubic block reduction). *The characteristic and minimal polynomials of U_X are both $T^2(T^2 - 108q)$. After adjoining r , its eigenvalues are $6r, -6r, 0, 0$, and the zero generalized eigenspace is one rank-two Jordan block. A normalized gauge regular in z gives*

$$\begin{aligned} z^2 \partial_z \tilde{y}_+ &= (6r + O(z^2)) \tilde{y}_+, \\ z^2 \partial_z \tilde{y}_- &= (-6r + O(z^2)) \tilde{y}_-, \\ z^2 \partial_z \begin{pmatrix} \tilde{y}_3 \\ \tilde{y}_4 \end{pmatrix} &= [N + zA_0 + z^2 A_1 + O(z^3)] \begin{pmatrix} \tilde{y}_3 \\ \tilde{y}_4 \end{pmatrix}, \end{aligned} \quad (2.15)$$

where

$$N = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad A_0 = \begin{pmatrix} -19/18 & 0 \\ 0 & 19/18 \end{pmatrix}, \quad A_1 = \begin{pmatrix} 0 & -14/(81r^2) \\ -8/81 & 0 \end{pmatrix}. \quad (2.16)$$

Proof. The matrix

$$C = \begin{pmatrix} 6r^3 & -6r^3 & 0 & -7r^2 \\ 7r^2 & 7r^2 & -2r^2 & 0 \\ 3r & -3r & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}, \quad \det C = -486r^5,$$

puts U_X into

$$C^{-1} U_X C = \begin{pmatrix} 6r & 0 & 0 & 0 \\ 0 & -6r & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Partition as $1 \mid 1 \mid 2$. At each positive order in z , the off-diagonal gauge coefficient is the unique solution of a Sylvester equation between blocks with disjoint scalar spectra. The gauge and inverse are regular in z ; explicitly, it has the normalized form

$$G(z) = I + \sum_{n \geq 1} G_n z^n, \quad G_n \text{ block-off-diagonal}. \quad (2.17)$$

To extract the coefficient used in the residue, write $y = CG(z)\tilde{y}$, $J = C^{-1}U_X C$ and $B = C^{-1}D_X C$. Label the blocks by $+, -, 0$. If the transformed coefficient is $J + z\hat{A}_0 + z^2\hat{A}_1 + \dots$, then

$$(J + zB)G - z^2 G' = G(J + z\hat{A}_0 + z^2\hat{A}_1 + \dots).$$

Thus $(\hat{A}_0)_{ii} = B_{ii}$, while the first Sylvester equations are

$$J_i(G_1)_{ij} - (G_1)_{ij}J_j = -B_{ij} \quad (i \neq j).$$

At the next order,

$$\hat{A}_1 = [J, G_2] + BG_1 - G_1 - G_1\hat{A}_0.$$

Since G_1, G_2 are block off-diagonal and $\hat{A}_0$ is block diagonal, the zero block is simply

$$A_1 = (\hat{A}_1)_{00} = B_{0+}(G_1)_{+0} + B_{0-}(G_1)_{-0} = \begin{pmatrix} 0 & -14/(81r^2) \\ -8/81 & 0 \end{pmatrix}.$$

This computes the needed coefficient without solving for G_2 , and proves (2.16). $\square$

For the zero block, $N = 2E_{12}$. With $S = \text{diag}(1, z)$,

$$R_X = 2E_{12} + A_0 - \frac{8}{81}E_{21} - \text{diag}(0, 1) = \begin{pmatrix} -19/18 & 2 \\ -8/81 & 1/18 \end{pmatrix}. \quad (2.18)$$

Its characteristic polynomial is $(T + 1/6)(T + 5/6)$. Thus the two exponent classes differ by $2/3 \notin \mathbf{Z}$, so the block is counted. Equivalently,

$$\boxed{\delta^\#(X) = \frac{4}{9}}, \quad I_{\text{exp}}(X) = 1. \quad (2.19)$$

From the small connection to the generic block. After inverting q , at the small quantum point the rank-two cluster is separated from the other eigenvalues. The cyclic-centralizer argument in Proposition 2.6 continues that cluster over the formal bulk ring and proves that its centered leading term remains nonzero and square-zero. The modified residue is then preserved by flatness. Thus the small calculation determines the exponent count at generic quantum parameters, for every smooth cubic threefold.

2.5 Vanishing on centers through dimension two

Proposition 2.9 (Vanishing on low-dimensional centers). *For every smooth projective Z of dimension at most two and every center summand specified by comparison data ω , $I_{\text{exp}}(Z; \omega) = 0$.*

Proof. Points and curves. A point has one rank-one block. For $\mathbf{P}^1$, the relation $p \star p = Qe^{t_p}$ gives two rank-one generic blocks.

Let C have genus $g \geq 1$. There is no nonconstant genus-zero map to C , so its even big quantum product is classical. With $\int_C p = 1$ and $a = 2 - 2g$, its centered even connection is

$$z\partial_z Y = \left[z^{-1}aE_{21} + \text{diag}(1/2, -1/2) \right] Y.$$

For $g = 1$, $N = 0$. For $g > 1$, the modification $S = \text{diag}(z, 1)$ gives residue $aE_{21} - \frac{1}{2}I$, of discriminant zero. Its two eigenvalues, and therefore its two exponent classes, are equal. Thus every curve has zero invariant.

Surfaces with nef canonical class. Let S be a surface with K_S nef. For a quantum-product term with input degree b , non-scalar Euler degree $a \geq 2$, curve class d , and nonunit even bulk degrees $e_\nu \geq 2$, the dimension axiom gives output degree

$$a + b - 2c_1(S) \cdot d + \sum_\nu (e_\nu - 2) \geq b + 2.$$

The centered Euler operator strictly raises ordinary degree. Its only block is the whole even cohomology, of rank $b_0 + b_2 + b_4 \geq 3$, so its invariant is zero.

Other surfaces. A minimal surface with non-nef canonical class is $\mathbf{P}^2$ or a ruled surface $\mathbf{P}_C(V)$ [4, Chapter VI]. The relation $H^{*3} = Q$ makes $\mathbf{P}^2$ generically semisimple. Formula (2.1) and the curve case give zero for ruled surfaces. Every nonminimal surface is obtained from a minimal one by point blowups; formula (2.2) adds only zero point terms. This proves intrinsic vanishing for every surface. Lemma 2.3 identifies every separate center summand with a faithful scalar extension of the corresponding intrinsic generic block, and the invariant is unchanged by that extension. $\square$

2.6 The projective endpoint and the contradiction

Formula (2.1) applied to $X \times \mathbf{P}^1 = \mathbf{P}_X(\mathcal{O}_X^{\oplus 2})$ gives

$$I_{\exp}(X \times \mathbf{P}^1) = 2. \quad (2.20)$$

On the other hand, over the generic numerical Novikov field $\Lambda = \text{Frac } \mathbf{C}[[Q]]$, $QH^\bullet(\mathbf{P}^4) = \Lambda[H]/(H^5 - Q)$. At $Q \neq 0$, multiplication by $5H$ has five distinct eigenvalues after algebraic closure. Thus

$$I_{\exp}(\mathbf{P}^4) = 0. \quad (2.21)$$

Proof of Theorem 1.1. Proposition 2.9 makes every actual center correction zero in a weak factorization of smooth projective fourfolds. Hence Theorem 2.2 makes $I_{\exp}$ birationally invariant. If $X \times \mathbf{P}^1$ were rational, then

$$2 = I_{\exp}(X \times \mathbf{P}^1) = I_{\exp}(\mathbf{P}^4) = 0,$$

a contradiction. □

3 Consequences

No new quantum calculation is needed in this section. We first apply the invariant to products of threefolds with $\mathbf{P}^1$, then transfer the stabilization theorem through Kuznetsov's flop, and finally combine it with known criteria for universal CH_0 -triviality.

Corollary 3.1 (Threefold criterion). *Let Y be a smooth projective complex threefold. If its generic even QDM contains a block counted by the exponent count defined in (2.12), then $Y \times \mathbf{P}^1$ is irrational. In particular, Y is irrational.*

Proof of Corollary 3.1. A counted block gives $I_{\exp}(Y) > 0$. The projective-bundle formula gives

$$I_{\exp}(Y \times \mathbf{P}^1) = 2I_{\exp}(Y) > 0.$$

Every center in projective weak factorization of fourfolds has dimension at most two, so Proposition 2.9 makes every correction zero. Theorem 2.2 therefore makes the invariant birationally invariant in dimension four. Since $I_{\exp}(\mathbf{P}^4) = 0$ by (2.21), $Y \times \mathbf{P}^1$ is irrational. Rationality of Y would imply rationality of this product, so Y is irrational as well. □

Kuznetsov's associated Pfaffian cubic and flop supply the next application. Tschinkel and Zhang study equivariant and twisted forms of this correspondence [22].

Corollary 3.2 (Irrationality after one stabilization for V_{14}). *For every smooth prime Fano threefold V of genus eight, $V \times \mathbf{P}^1$ is irrational.*

Proof of Corollary 3.2. Kuznetsov's flop gives a birational map between rank-two projective bundles $\mathbf{P}_V(U)$ and $\mathbf{P}_X(E^*)$ over V and its associated Pfaffian cubic threefold X [18, Theorems 2.17–2.18]. Each projective bundle is birational to the product of its base with $\mathbf{P}^1$. Hence $V \times \mathbf{P}^1$ is birational to $X \times \mathbf{P}^1$, which is irrational by Theorem 1.1. □

Corollary 3.3 (Separation in codimension at most three). *There is a nonempty countable union of subvarieties of codimension at most three in the moduli space of smooth cubic threefolds such that, for every member X , both X and $X \times \mathbf{P}^1$ are universally CH_0 -trivial, while $X \times \mathbf{P}^1$ is irrational.*

Proof of Corollary 3.3. Voisin produces a nonempty countable union of subvarieties of codimension at most three in the moduli space of smooth cubic threefolds along which the primitive minimal class $\Theta^4/4!$ is algebraic, hence along which CH_0 is universally trivial [23, Theorem 4.5 and Lemma 4.6]. The projective-bundle formula passes universal CH_0 -triviality to the product, and Theorem 1.1 supplies its irrationality. $\square$

Corollary 3.4 (Fermat cubic). *Let*

$$X = \{x_1^3 + x_2^3 + x_3^3 + x_4^3 + x_5^3 = 0\} \subset \mathbf{P}^4.$$

Then the fourfold $X \times \mathbf{P}^1$ is universally CH_0 -trivial and irrational.

Proof of Corollary 3.4. The Fermat equation is a sum of five forms in separated single variables, so [9, Theorem 2.8] gives universal CH_0 -triviality; the projective-bundle formula passes it to the product. Irrationality follows from Theorem 1.1. $\square$

Corollary 3.5 (Coprime-degree cubics). *Let X be one of the cubic threefolds of [24, Theorem 3.3]. Then $X \times \mathbf{P}^1$ admits unirational parametrizations of coprime degrees, is universally CH_0 -trivial, and is irrational.*

Proof of Corollary 3.5. The cubics of [24, Theorem 3.3] carry unirational parametrizations of degrees two and three, and [24, Corollary 3.5] keeps both degrees on $X \times \mathbf{P}^1$. Parametrizations of coprime degrees give a decomposition of the diagonal and hence universal CH_0 -triviality [24, Section 1]. Theorem 1.1 gives irrationality. $\square$

In dimension five, threefold centers can contribute to weak factorization. The next section retains those contributions and places the blowup formulas in the additive Grothendieck group.

4 Additive and spectral extensions

The blowup formulas also define an additive invariant on the Grothendieck group of varieties. We first retain the squared exponent gaps instead of only counting them. The canonical modified lattice is essential here: arbitrary separate integer shifts of the two exponents can change their squared difference.

Lemma 4.1 (A fixed target for the residue spectrum). *For every counted generic block, $\delta^\#$ belongs to*

$$\mathcal{D} = \mathbf{C} \setminus \{n^2 : n \in \mathbf{Z}\}.$$

Its value is preserved by the regular comparisons used in the blowup and projective-bundle formulas.

Proof. The finite jet defining the residue is over a finite algebraic extension of the reduced generic coefficient field K_Y : spectral projectors and the finitely many Sylvester equations needed for this jet require only algebraic scalar extension. Proposition 2.6 makes its discriminant constant in every even bulk direction. On the reduced source, a divisor direction acts by $X_d \mapsto (D \cdot d)X_d$, while the other bulk directions are ordinary partial derivatives. Thus the discriminant is killed by all logarithmic derivations in the numerical curve and remaining bulk variables.

The common constants of K_Y under these derivations are $\mathbf{C}$. Here is a coefficient argument that also applies to the completed ring. Write a constant fraction as f/g . Simultaneous formal rescaling of the monomials gives $f(e^t x)g(x) = f(x)g(e^t x)$, by uniqueness

of formal Taylor expansion. At each bounded ample-plus-bulk degree only finitely many monomials occur. Their distinct exponential characters are linearly independent, by the Vandermonde argument in Lemma 2.3. Comparing the coefficient of each monomial therefore gives $f_a g_b = g_a f_b$ for every pair of indices a, b . Choose b with $g_b \neq 0$; then $f = (f_b/g_b)g$. The same constant field holds in an algebraic extension: differentiating the monic minimal polynomial of an algebraic constant gives a polynomial of smaller degree vanishing at it, so every coefficient of the minimal polynomial is constant. Since $\mathbf{C}$ is algebraically closed, the algebraic constant lies in $\mathbf{C}$. The exponent criterion excludes exactly the integer squares.

A regular comparison maps the centered leading operator and its image line to their counterparts. It therefore maps $E^\sharp$ isomorphically to the corresponding modified lattice. On reduction modulo z , the residues are conjugate. A common scalar shift of the residue also leaves $(\mathrm{tr} R)^2 - 4\det R$ unchanged. These are the transformations supplied by the comparisons; no separate change of the two exponent representatives is needed. The faithful coefficient maps fix $\mathbf{C}$, so they preserve the same numerical value in $\mathcal{D}$. $\square$

Let $M_{\mathrm{disc}} = \bigoplus_{\delta \in \mathcal{D}} \mathbf{N}e_\delta$, with group completion $G_{\mathrm{disc}} = \bigoplus_{\delta \in \mathcal{D}} \mathbf{Z}e_\delta$. For a connected smooth projective variety put

$$\mathfrak{S}(Y) = \sum_{E \text{ counted}} e_{\delta^\sharp(E)} \in M_{\mathrm{disc}}.$$

Extend by sums over connected components and put $\mathfrak{S}(\emptyset) = 0$. The augmentation $\epsilon : G_{\mathrm{disc}} \rightarrow \mathbf{Z}$, $e_\delta \mapsto 1$, recovers I_{exp} .

Proposition 4.2 (Intrinsic operation formulas). *Let Y be smooth projective over $\mathbf{C}$. For a smooth closed center $Z \subset Y$ of codimension $c \geq 1$ and a rank- r vector bundle V on Y ,*

$$\mathfrak{S}(\mathrm{Bl}_Z Y) = \mathfrak{S}(Y) + (c-1)\mathfrak{S}(Z), \quad \mathfrak{S}(\mathbf{P}_Y(V)) = r\mathfrak{S}(Y).$$

The same formulas hold for I_{exp} . For centers of differing component dimensions, apply the formula separately to each component.

Proof. Lemma 2.3 identifies every center occurrence with a faithful extension of its intrinsic module in every dimension. Lemma 4.1 preserves each summand's spectrum. Independent unit coordinates separate the summands generically, as in Proposition 2.4. Consequently their spectra add. The projective-bundle comparison gives r copies in the same way. Augmentation proves the scalar formulas. Disjoint centers may be blown up successively, giving the last assertion. $\square$

Theorem 4.3 (Additive Grothendieck-group invariants). *There are unique additive homomorphisms*

$$\mathcal{S} : K_0(\mathrm{Var}_{\mathbf{C}})/(\mathbb{L} - 1) \longrightarrow G_{\mathrm{disc}}, \quad \mathcal{I} = \epsilon\mathcal{S} : K_0(\mathrm{Var}_{\mathbf{C}})/(\mathbb{L} - 1) \longrightarrow \mathbf{Z}$$

agreeing with $\mathfrak{S}$ and I_{exp} on smooth projective varieties. Here $\mathbb{L} = [\mathbf{A}^1]$, $\mathcal{I}(1) = \mathcal{I}(\mathbb{L}) = 0$, and for a smooth cubic threefold X , $\mathcal{S}([X]) = e_{4/9}$ and $\mathcal{I}([X]) = 1$.

Proof. Bittner's presentation of the abelian group uses smooth projective generators, the empty class, and the blowup relation [2, Theorem 3.1]. Since the exceptional divisor is $\mathbf{P}_Z(N_{Z/Y})$, Proposition 4.2 gives

$$\mathfrak{S}(\mathrm{Bl}_Z Y) - \mathfrak{S}(\mathbf{P}_Z(N_{Z/Y})) = \mathfrak{S}(Y) - \mathfrak{S}(Z)$$

in G_{disc} . Thus the assignment extends uniquely and additively. For a smooth projective Y , the scissor relation and the bundle formula give

$$[Y \times \mathbf{P}^1] = [Y] + \mathbb{L}[Y], \quad \mathcal{S}(\mathbb{L}[Y]) = \mathfrak{S}(Y \times \mathbf{P}^1) - \mathfrak{S}(Y) = \mathfrak{S}(Y).$$

Smooth projective classes generate the whole group, so $\mathcal{S}((\mathbb{L} - 1)a) = 0$ for every a . This proves the factorization through the indicated quotient, and augmentation gives $\mathcal{I}$. Points have no counted block, while the cubic has the single block of discriminant $4/9$ computed in Section 2.4. $\square$

These are additive group invariants. In particular, $\mathcal{I}$ is not multiplicative: its value at 1 is zero but its value at a cubic is one. The identity $\mathcal{I}(\mathbb{L}a) = \mathcal{I}(a)$ does not make $\mathcal{I}$ a stable-birational invariant.

For $n \geq 1$, the relation $H^{n+1} = q$ in small quantum cohomology shows that Euler multiplication on $\mathbf{P}^n$ has distinct eigenvalues at $q \neq 0$. It remains semisimple at the generic big point. Thus $\mathfrak{S}(\mathbf{P}^n) = 0$ for every n , including $n = 0$. For example, embed a cubic threefold $X \subset \mathbf{P}^4 \subset \mathbf{P}^5$. Then $\mathcal{I}([\text{Bl}_X \mathbf{P}^5]) = 1$, whereas $\mathcal{I}([\mathbf{P}^5]) = 0$. Thus birational invariance already fails in dimension five.

Proposition 4.4 (Equal Hodge diamonds, different invariant values). *Let X be a smooth cubic threefold and $p \in X$. Let $C \subset \mathbf{P}^1 \times \mathbf{P}^1 \subset \mathbf{P}^3$ be a smooth curve of bidegree $(2, 6)$. Then $\text{Bl}_p X$ and $\text{Bl}_C \mathbf{P}^3$ have the same Hodge diamond, but their invariant values are respectively one and zero. Consequently $\mathcal{I}$ does not factor through the Hodge–Deligne polynomial.*

Proof. The linear system $\mathcal{O}(2, 6)$ contains smooth members, and adjunction gives genus $(2 - 1)(6 - 1) = 5$. For the cubic, $c(T_X) = (1 + H)^5/(1 + 3H)$ gives Euler characteristic -6 . Weak Lefschetz and $K_X = \mathcal{O}_X(-2)$ then give $h^{1,1} = h^{2,2} = 1$, $h^{2,1} = h^{1,2} = 5$, and $h^{3,0} = h^{0,3} = 0$. The Hodge blowup formula shows that both displayed blowups have, besides $h^{0,0} = h^{3,3} = 1$, precisely

$$h^{1,1} = h^{2,2} = 2, \quad h^{2,1} = h^{1,2} = 5.$$

Their Hodge–Deligne polynomials therefore agree. The intrinsic blowup formula adds only point or curve contributions, all zero, so their invariant values equal those of X and $\mathbf{P}^3$, namely one and zero. $\square$

Corollary 4.5 (Contributions in any rationalizing factorization). *Let X be a smooth cubic threefold. In any projective weak factorization of a rationalization $X \times \mathbf{P}^m \dashrightarrow \mathbf{P}^{m+3}$, orient the arrows from the product to projective space and take connected centers. Then*

$$\sum_{\text{blowdowns}} (c_Z - 1)\mathfrak{S}(Z) - \sum_{\text{blowups}} (c_Z - 1)\mathfrak{S}(Z) = (m + 1)e_{4/9}.$$

For $m = 2$, only threefold centers of codimension two can contribute; their net spectrum is $3e_{4/9}$, and their net contribution to I_{exp} is three.

Proof. The initial spectrum is $(m + 1)e_{4/9}$; projective space has no counted blocks and hence spectrum zero. Telescope the intrinsic blowup formulas. In dimension five, every nontrivial center has dimension at most three, and centers of dimension at most two contribute zero. The remaining centers have codimension two. Applying ϵ gives the scalar identity. This counts contributions with multiplicity, not distinct centers, and does not identify a center as a cubic threefold. $\square$

Limits of the construction. Smooth complex cubic threefolds form one connected deformation family. Unmarked deformation-invariant Gromov–Witten data are consequently the same throughout that family, whereas it contains both stably rational cubics [21] and very general stably irrational cubics [10]. Increasing the collection of such quantum data cannot by itself completely detect stable rationality within the family.

The rank-two pairing argument also has a specific boundary. In rank three, let E_{ij} denote the matrix units and put

$$N = E_{12} + E_{23}, \quad P = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad A_0 = E_{21} - E_{32}.$$

Then

$$N^T P = P N = E_{23} + E_{32}, \quad P A_0 = E_{21} - E_{12} = -A_0^T P.$$

Thus the constant pairing is horizontal for $A(z) = N/z + A_0$, but $A_0 e_1 = e_2 \notin \ker N$. Pairing horizontality alone does not preserve the kernel flag. For a cyclic Jordan leading term and $S = \text{diag}(1, z, \dots, z^{r-1})$, regularity of the modified connection requires $(A_k)_{ij} = 0$ whenever $i - j > k$, for $k \geq 0$. Additional geometric filtration compatibility would therefore be needed for that higher-rank modification.

AI assistance disclosure

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Numbered statements are cited from the exact versions listed here. Where both a journal and a preprint version appear, the numbering is that of the preprint.

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